Unit NCIR

SKU:U028











Description

Unit NCIR is a single-point infrared temperature measurement sensor. It features a built-in infrared sensor **MLX90614**, capable of measuring the surface temperature of the human body or other objects.

Unlike most contact-type sensors, this sensor detects temperature by measuring infrared light emitted from distant objects, without physical contact. This gives it a wider temperature measurement range: -70 °C to + 380 °C. With a field of view of 90°, it can quickly and conveniently measure the average temperature at a certain position.

This Unit connects to M5Core via PORT A I2C (0x5A).

Features

- MLX90614ESF-BAA
- Object and ambient temperature measurement
- Development Platform: Arduino, UIFlow (Blockly, Python)
- 2 x LEGO-compatible holes

Includes

- 1 x Unit NCIR

- 1 x HY2.0-4P Grove Cable (20cm)

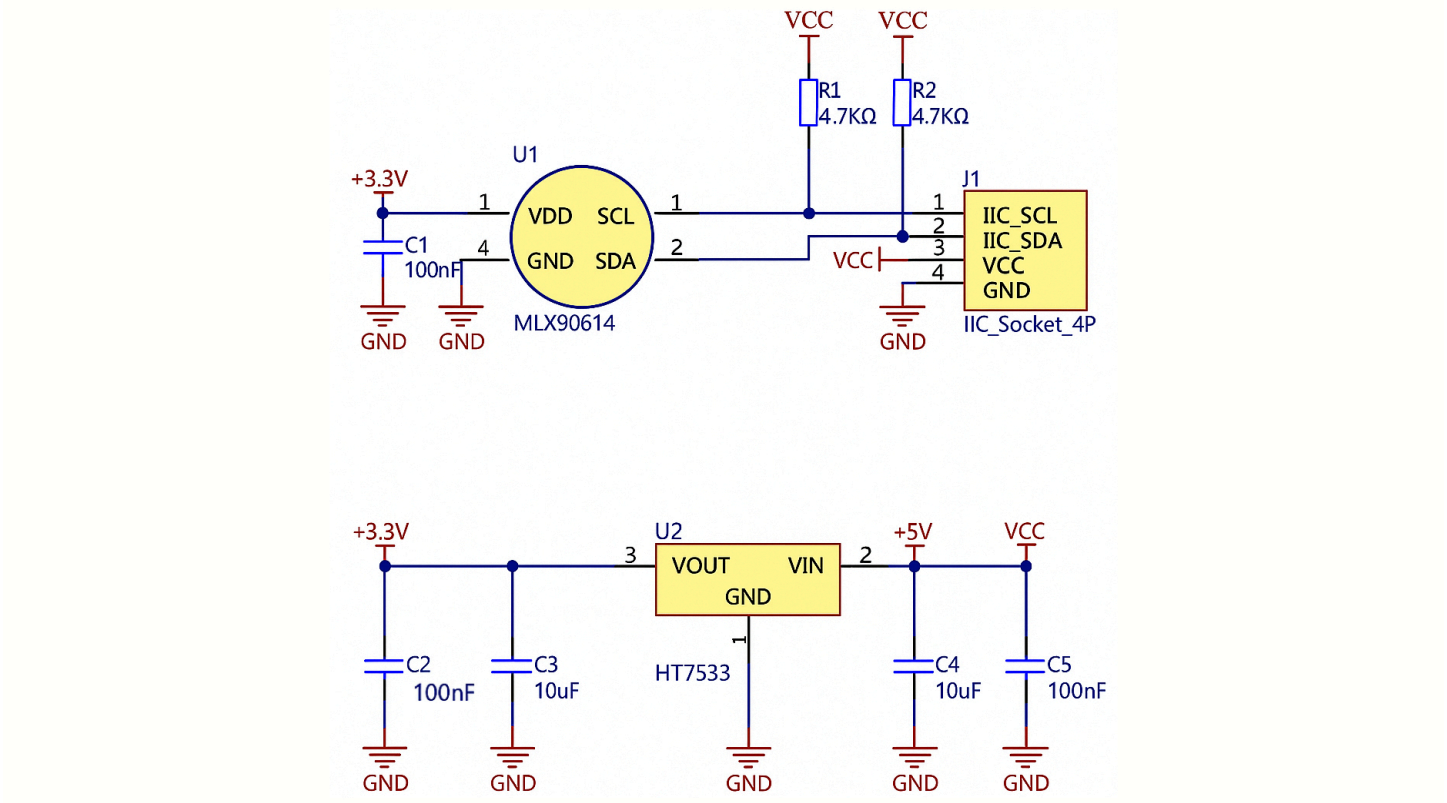
Applications

- Human body temperature measurement
- Object (biological) motion detection

Specifications

Specification	Parameter
Communication Interface	I2C Communication @ 0x5A, Interface Speed: 100Kbps
Object Temperature	-70 °C ~ 380 °C
Ambient Temperature	-40 °C ~ 125 °C
Accuracy	±0.5 °C
Field of View	90°
Product Size	32.0 x 24.0 x 8.6mm
Product Weight	4.8g
Package Size	138.0 x 93.0 x 13.0mm
Gross Weight	10.0g

Schematics

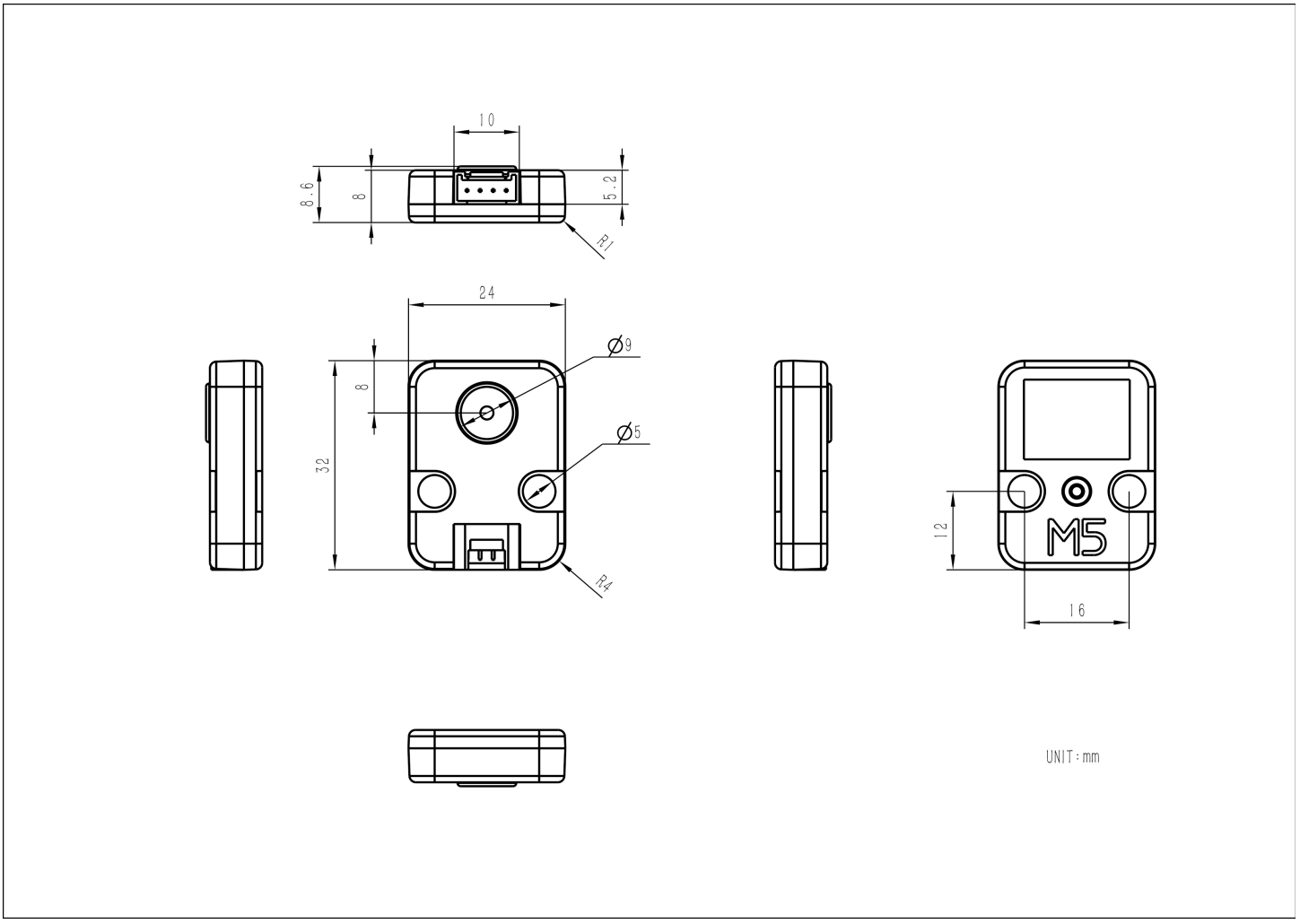


PinMap

Unit NCIR

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Model Size



Structure

- Unit NCIR Structure Files

Datasheets

- MLX90614 Datasheet

Softwares

Arduino

- Unit NCIR Example

UiFlow2

- Unit NCIR UiFlow2 Docs

Home Assistant

- Unit NCIR Home Assistant Integration

EasyLoader

Easyloader	Download	Note
Unit NCIR Test Easyloader	download	/

Video

- Display current detected temperature value on screen.

NCIR_UNIT.mp4